

**Interdisciplinary Project - 1BPRJ208 Synopsis****Academic year- 2025-26**

Name of Guide: Dr. Chetan A S and Dr. Nagabhushan S V		Project Batch/Group Number: 01	Section: AIML - 01
Name Of Student (Block Letters)	USN (Block Letters)	Email-Id	Contact No.
CHETAN M D	1BY25AI061	25ug1byai359@bmsit.in	8147366176
AYUSH KUMAR YADAV	1BY25AI043	25ug1byai288@bmsit.in	7803077193
GEETIKA DEVI	1TD25AI030	25ug1byai004@bmsit.in	9541032417
VAISHNAVI G	1BY25AI265	25ug1byai197@bmsit.in	9494734802

Project Title:	House Price Prediction using Newton–Raphson Method
Abstract	This project aims to predict house prices using a simple machine learning approach based on linear regression. The model establishes a relationship between house size and price. To improve accuracy and efficiency, the Newton–Raphson method is used as an optimization technique to minimize the prediction error. Unlike traditional methods such as gradient descent, Newton–Raphson uses both first and second derivatives, allowing faster convergence to the optimal solution. This project demonstrates how basic mathematical concepts like derivatives and error minimization can be applied to solve real-world problems in a simple and beginner-friendly manner.
Introduction	House price prediction is an important application in real estate and financial decision-making, where accurate estimation of property value plays a key role for buyers, sellers, and investors. Various studies have explored the use of machine learning techniques, especially linear regression, to model the relationship between house features and price. In numerical optimization research, the Newton–Raphson method has been widely used to solve equations efficiently by utilizing both first and second derivatives. Compared to gradient-based methods, it provides faster convergence to the optimal solution, making it a powerful alternative for parameter optimization. In this project, a simple linear regression model is used to predict house prices based on size. The Newton–Raphson method is applied to minimize the error, providing faster and more efficient learning while keeping the model easy to understand for beginners.



BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

Yelahanka, Bengaluru. 560064

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Existing System	<table> <tr> <th>Paper Details</th><th>Observations</th><th>Gap Identified</th><th>References Link</th></tr> <tr> <td> Title: A Study on Linear Regression for House Price Prediction Authors: Generic academic study (survey-based reference) </td><td>Gradient descent requires multiple iterations</td><td>Need for faster convergence</td><td>en.wikipedia.org/wiki/Linear_Regression</td></tr> <tr> <td rowspan="2"> Title: Application of Newton–Raphson Method in Optimization Problems Authors: Numerical methods research study </td><td>Convergence can be slow depending on learning rate.</td><td>More efficient optimization method</td><td>en.wikipedia.org/wiki/Newton%27s_method</td></tr> <tr> <td>Requires tuning of parameters</td><td>Simpler implementation for small datasets</td><td></td></tr> </table>	Paper Details	Observations	Gap Identified	References Link	Title: A Study on Linear Regression for House Price Prediction Authors: Generic academic study (survey-based reference)	Gradient descent requires multiple iterations	Need for faster convergence	en.wikipedia.org/wiki/Linear_Regression	Title: Application of Newton–Raphson Method in Optimization Problems Authors: Numerical methods research study	Convergence can be slow depending on learning rate.	More efficient optimization method	en.wikipedia.org/wiki/Newton%27s_method	Requires tuning of parameters	Simpler implementation for small datasets			
Paper Details	Observations	Gap Identified	References Link															
Title: A Study on Linear Regression for House Price Prediction Authors: Generic academic study (survey-based reference)	Gradient descent requires multiple iterations	Need for faster convergence	en.wikipedia.org/wiki/Linear_Regression															
Title: Application of Newton–Raphson Method in Optimization Problems Authors: Numerical methods research study	Convergence can be slow depending on learning rate.	More efficient optimization method	en.wikipedia.org/wiki/Newton%27s_method															
	Requires tuning of parameters	Simpler implementation for small datasets																
Problem Statement		To design a simple predictive model that estimates house prices using the Newton–Raphson method.																
Objectives		- To understand linear regression fundamentals - To implement a basic machine learning model - To minimize error using mathematical techniques - To apply Newton–Raphson optimization - To compare efficiency with traditional methods																
Proposed System		The proposed system uses a linear regression model where house price is predicted based on size. The model parameters (weight and bias) are optimized using the Newton–Raphson method instead of gradient descent. This approach improves convergence speed and demonstrates efficient learning using fewer iterations.																
Methodology		Step 1: Define Model $y = wx + b$ Where: - x = house size - y = predicted price - w = weight - b = bias Step 2: Prediction Equation $y_{\text{predicted}} = wx + b$																



	Step 3: Error Function $\text{Error} = \sum (y_{\text{actual}} - y_{\text{predicted}})^2$ Step 4: Gradient (First Derivative) $d(\text{Error})/dw = -2\sum x(y - y_{\text{predicted}})$ Step 5: Hessian (Second Derivative) $d^2(\text{Error})/dw^2 = 2\sum x^2$ Step 6: Newton–Raphson Update $w_{\text{new}} = w_{\text{old}} - \text{Gradient/Hessian}$ Step 7: Iteration Repeat the update until error becomes minimum.
System Requirement Specifications • Hardware Requirements • Software Requirement	Hardware Requirements • Basic computer system • Minimum 4GB RAM Software Requirements • Python • NumPy library • VS Code / Jupyter Notebook
Applications of the Project	• Real estate price prediction • Property valuation systems • Investment decision support • Financial planning tools
Conclusion	• The project successfully demonstrates house price prediction using a simple linear regression model. • The relationship between house size and price is effectively modeled using a mathematical equation. • The Newton–Raphson method is applied to optimize the model parameters efficiently. • It uses both first and second derivatives, resulting in faster convergence compared to traditional methods like gradient descent. • The error minimization process helps improve prediction accuracy step by step. • The project provides a clear understanding of basic machine learning concepts such as prediction, error, and optimization. • It shows how mathematical techniques can be applied to solve real-world problems in a simple and effective manner.
References : (IEEE format)	[1] T. Hastie, R. Tibshirani, and J. Friedman, <i>The Elements of Statistical Learning</i>, Springer, 2009. [2] S. Boyd and L. Vandenberghe, <i>Convex Optimization</i>, Cambridge University Press, 2004. [3] Python Software Foundation, “Python Documentation,” Available: https://docs.python.org



BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

Yelahanka, Bengaluru. 560064

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

	[4] “Linear Regression,” Wikipedia, Available: https://en.wikipedia.org/wiki/Linear_regression [5] “Newton’s Method,” Wikipedia, Available: https://en.wikipedia.org/wiki/Newton%27s_method
Mention any two SDG goals that are mapped with this project	1. SDG 11: Sustainable Cities and Communities 2. SDG 9: Industry, Innovation and Infrastructure

Signature of Guide

Signature of Division Head

Signature of HOD